
LITERATURE SURVEY

MedTruth AI: An Evidence-Grounded Medical Report Assistant

DOMAIN

Healthcare

10 Research Papers Reviewed

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01

ChatGPT Makes Medicine Easy to Swallow: An Exploratory Case Study on Simplified Radiology Reports

Katharina Jeblick et al.

KEY FINDINGS

- Evaluates ChatGPT for simplifying complex radiology reports for patients.
- Found that ChatGPT can make medical information easier for patients to understand.
- Most evaluated reports were considered factually correct and complete.
- Some errors, hallucinations, and missing information were identified.
- Highlights the need for expert review of AI-generated medical content.
- Supports the use of AI assistants like MedTruth AI for understandable healthcare communication.

02

Generative Artificial Intelligence to Transform Inpatient Discharge Summaries to Patient-Friendly Language and Format

Jonah Zaretsky, Jeong Min Kim, Samuel Baskharoun, Yunan Zhao, Jonathan Austrian, Yindalon Aphinyanaphongs, Ravi Gupta, Saul B. Blecker, Jonah Feldman

KEY FINDINGS

- Uses GPT-4 to convert complex hospital discharge summaries into patient-friendly language.
- Tested on 50 real inpatient discharge summaries from NYU Langone Health.
- AI-generated summaries were shorter and easier to understand than original medical notes.
- Readability improved from approximately 11th-grade to 6th-grade level.
- Omissions and hallucinations caused some accuracy and safety concerns.
- Supports MedTruth AI for simplifying medical records while requiring clinical review for safety.

03

Explainability, Transparency and Black Box Challenges of AI in Radiology: Impact on Patient Care in Cardiovascular Radiology

Ahmed Marey, Parisa Arjmand, Ameerh Dana Sabe Alerab, Mohammad Javad Eslami, Abdelrahman M. Saad, Nicole Sanchez, Muhammad Umair

KEY FINDINGS

- Reviews the use of AI in cardiovascular imaging and its impact on patient care.
- Focuses on the black-box problem, where AI decisions are difficult for clinicians to understand.
- Lack of transparency can reduce clinical trust and create ethical and legal concerns.
- Explainable AI techniques such as Grad-CAM can show which image regions influenced AI decisions.
- Recommends interpretable and hybrid AI models, along with user-friendly interfaces.
- These approaches can make AI safer and more trustworthy for applications such as MedTruth AI.

04

Accuracy, Consistency, and Hallucination of Large Language Models When Analyzing Unstructured Clinical Notes in Electronic Medical Records

Savyasachi V. Shah, BDS, MPH

KEY FINDINGS

- Examines the use of LLMs for extracting information from unstructured clinical notes.
- Highlights the potential of LLMs to process large amounts of medical-record text efficiently.
- LLM performance depends strongly on prompt quality, with detailed prompts producing better results.
- Identifies problems with hallucinations, negations, ambiguous terms, and inconsistent outputs.
- Combining LLMs with rule-based NLP methods and human oversight may improve reliability.
- Relevant to MedTruth AI, where reliable extraction and summarization of patient medical records are important.

05

Improving Reliability and Explainability of Medical Question Answering through Atomic Fact Checking in Retrieval-Augmented LLMs

Juraj Vladika, Annika Domres, Mai Nguyen, Rebecca Moser, Jana Nano, Felix Busch, Lisa C. Adams, Keno K. Bressemer, Denise Bernhardt, Stephanie E. Combs, Kai J. Borm, Florian Matthes, Jan C. Peeken

KEY FINDINGS

- Proposes an atomic fact-checking framework to improve medical LLM answers.
- Divides generated answers into individual verifiable facts.
- Each fact is checked against authoritative medical guidelines using RAG.
- Incorrect or hallucinated facts are identified, corrected, and linked to supporting sources.
- The approach improved answer quality by up to 40% and improved factual reliability.
- Highly relevant to MedTruth AI, helping provide trustworthy, explainable, evidence-based healthcare assistance.

06

Retrieval Augmented Generation for Large Language Models in Healthcare: A Systematic Review

Lameck Mbangula Amugongo, Pietro Mascheroni, Steven Brooks, Stefan Doering, Jan Seidel

KEY FINDINGS

- Systematically reviews how RAG is being used with LLMs in healthcare.
- Examines retrieval, augmentation, generation methods, datasets, and evaluation approaches.
- Shows that RAG can ground LLM responses using external medical knowledge and reduce hallucinations.
- Covers Naive RAG, Advanced RAG, and Modular RAG approaches used in healthcare.
- Highlights the importance of reliable evidence, standardized evaluation, privacy, safety, and ethics.
- Relevant to MedTruth AI as it supports using evidence-grounded RAG for more reliable healthcare responses.

07

Grounding Large Language Models in Clinical Evidence: A Retrieval-Augmented Generation System for Querying UK NICE Clinical Guidelines

Matthew Lewis, Samuel Thio, Amy Roberts, Catherine Siju, Whoasif Mukit, Rebecca Kuruvilla, Zhangshu Joshua Jiang, Niko Möller-Grell, Aditya Borakati, Richard JB Dobson, Spiros Denaxas

KEY FINDINGS

- Develops a RAG-based AI system for accessing NICE clinical guidelines.
- Retrieves relevant medical evidence before generating an answer.
- Hybrid search combines keyword-based and semantic retrieval for better accuracy.
- RAG greatly improves answer faithfulness and reduces hallucinations.
- Clinical experts found the system highly accurate and safer than non-RAG models.
- Highly relevant to MedTruth AI for evidence-grounded and trustworthy healthcare answers.

08

Large Language Model Answers Medical Questions About Standard Pathology Reports

Anqi Wang, Jieli Zhou, Peng Zhang, Haotian Cao, Hongyi Xin, Xinyun Xu, Haiyang Zhou

KEY FINDINGS

- Evaluates ChatGPT for answering questions from colorectal cancer pathology reports.
- An OCR-based system converts uploaded pathology report images into text for AI analysis.
- AI performed well in explaining pathology type, stage, and immunohistochemical results.
- Less reliable for treatment, prognosis, and follow-up recommendations.
- Highlights the potential of LLMs to improve patient understanding and doctor-patient communication.
- Relevant to MedTruth AI as it supports medical-record interpretation while showing the need for RAG-based evidence grounding.

09

The Role of Artificial Intelligence in Healthcare: A Structured Literature Review

Silvana Secinaro, Davide Calandra, Aurelio Secinaro, Vivek Muthurangu, Paolo Biancone

KEY FINDINGS

- Presents a structured literature review of Artificial Intelligence applications in healthcare.
- Analyzed 288 peer-reviewed papers from the Scopus database using qualitative and quantitative methods.
- Identifies major AI applications including predictive medicine, patient data analysis, diagnostics, and clinical decision-making.
- Highlights how AI can help professionals extract useful information from large datasets for faster, informed decisions.
- Emphasizes challenges such as data quality, ethics, data governance, and healthcare workforce skills.
- Relevant to MedTruth AI as it supports using AI for reliable healthcare decision support with data quality and transparency.

10

MERA: Medical Electronic Records Assistant

Ahmed Ibrahim, Abdullah Khalili, Maryam Arabi, Aamenah Sattar, Abdullah Hosseini, Ahmed Serag

KEY FINDINGS

- Presents MERA, a RAG-based AI system for analyzing medical records.
- Provides medical question answering, report summarization, and similarity search.
- RAG retrieves relevant medical information before generating responses, helping reduce hallucinations.
- Uses a vector database and cross-encoder to retrieve and rank clinically relevant information.
- Evaluated using synthetic medical reports and de-identified MIMIC-IV EHR data, showing strong performance.
- Highly relevant to MedTruth AI as it demonstrates evidence-grounded medical record analysis and explainable clinical support.